SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
SOFTWARE ENGINEERING DEPARTMENT

COURSE INFORMATION SHEET
(For Lab Based Course)

Session:	Fall-2022
Course Title:	Web Engineering
Course Code:	SWE-315L
Credit Hours:	1
Semester:	6th
Pre-Requisites:	Programming Fundamentals
Instructor Name:	Ms. Fatima Waseem, Ms. Sitwat Ashraf
Email and Contact Information:	fwsatti@ssuet.edu.pk , sitwat.ashraf@ssuet.edu.pk 0316-5310538, 03333809034
WhatsApp Group	Yes
Office Hours:	09:00 AM To 5:00 PM
Mode of Teaching:	Synchronous/Asynchronous/ Hybrid/Blended

COURSE OBJECTIVE:

The goal of the course is to give knowledge about web engineering concepts, principles process, methods and models. How to model design for web applications and what models' perspective are useful for construction. Familiarize with Web application development software tools and environments currently available on the market. Build Web Applications that are scalable flexible to modify and easy to manage.

COURSE OUTLINE:

Web programming languages, Design principles of Web based applications, Web platform constraints, Web standards, Responsive Web Design, Web Applications, Browser/Server Communication, Storage Tier, Cookies and Sessions, Input Validation, Full stack state management, Web App Security, Large scale applications, Performance of Web Applications, Enabling Technologies for adoption of cloud services and future Data Centers, Web Testing and Web Maintenance.

COURSE OUTCOMES:

CO No.	Course Outcomes (COs)
1	Understand the processes, concepts and standards related to the web engineering.
2	Produce static and responsive designs for web application.
3	Develop dynamic web application.
4	Apply techniques to resolve security threats in web applications.
5	Apply web standards for web application development
6	Analyze and solve small-scale web engineering problems

7	Use modern engineering tools necessary for web engineering practice

GRADING POLICY:

Assessment Tools	Percentage
Lab Manual	30%
Viva/Demonstration	30%
Lab Exam	40%
TOTAL	**100%**

Recommended Book:

- **Web Engineering the Discipline of Systematic Development of Web Applications** by Gerti Kappel, Birgit Proll, Siegfried Reich, Werner R. ISBN 13: 9780470015544

Reference Books:

- **Web Engineering: Modelling and Implementing Web Applications (Human–Computer Interaction Series)** 2014 Edition by Rossi ISBN 978-1-84628-923-1
- **Current Trends in Web Engineering** ICWE 2018 International Workshops, 2018

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LAB PLAN

Course Title: **Web Engineering**
Course Code: **SWE-315**

Week No.	Lab Date	Objective	Required Reading
1	10-10-2022 to 14-10-2022	HTML, CSS	Lab Manual Pages 1 - 8
2	17-10-2022 to 21-10-2022	Basic JavaScript	Lab Manual Pages 9 – 14
3	24-10-2022 to 28-10-2022	Document Object Model [DOM]	Lab Manual Pages 15 - 18
4	31-10-2022 to 04-11-2022	Browser Object Model [BOM]	Lab Manual Pages 19 - 22
5	07-11-2022 to 11-11-2022	Intermediate JavaScript	Lab Manual Pages 23 – 27
6	14-11-2022 to 18-11-2022	Bootstrap/SASS	Lab Manual Pages 28 – 32
7	21-11-2022 to 25-11-2022	ES5\ES6\ESNEXT\Typescript	Lab Manual Pages 33 – 37
8	28-11-2022 to 02-12-2022	JavaScript- The Good Parts	Lab Manual Pages 38 – 40
9	**Mid Term Examination (05-12-2022 to 10-12-2022)**		
10	12-12-2022 to 16-12-2022	Bundling / Webpack	Lab Manual Pages 41 – 43
11	19-12-2022 to 23-12-2022	React Class / Functional	Lab Manual Pages 44 – 47
12	26-12-2022 to 30-12-2022	React / Redux /Context	Lab Manual Pages 48 – 52
13	02-01-2023 to 06-01-2023	React Router	Lab Manual Pages 53 – 59
14	09-01-2023 to	Formik Forms + Material UI	Lab Manual

	13-01-2023		Pages 60 – 62
15	16-01-2023 to 20-01-2023	E2E CRUD App Development using MongoDB, NodeJs	Lab Manual Pages 63 – 70
16	**Lab Examination (23-01-2023 to 27-01-2023)**		